Microscopic Traffic Simulation of a Signalised Intersection in Ghent, Belgium

STUDY AREA

R40–N430
Neuseplein–Tolpoort, Ghent
Signalised intersection K1

SOFTWARE AND DATA

PTV Vissim 2026 SP10
Student Version
AWV counts, 9 February 2023
AWV signal plan V010049 v17

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ABOVE

The study area: the signalised R40–N430 intersection at Neuseplein–Tolpoort, Ghent.

RIGHT

The same junction as reconstructed in PTV Vissim, with trams on the R40.



DEMAND STRESS TESTING, MULTIMODAL OPERATIONS AND TRAM PRIORITY

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1 CHAPTER

Project Overview

This independent project examines traffic operations at a signalised intersection in Ghent, Belgium, using microscopic traffic simulation in PTV Vissim. The model was developed to investigate how the intersection responds to increased motor traffic demand and the introduction of tram operations with priority control.

The network was reconstructed to represent the existing road layout, including motor vehicle movements and bicycle infrastructure. Traffic demand was derived from turning-movement counts provided by the Flemish Agency for Roads and Traffic (Agentschap Wegen en Verkeer, AWV), while the baseline signal control was implemented using an official signal timing plan.

The modelling process covers network reconstruction, demand coding, signal and conflict modelling, and a comparison of simulated queues with available field observations. Demand stress tests were subsequently conducted to examine changes in intersection performance under higher traffic volumes. A separate scenario introduces tram movements and simplified priority control to assess their operational effects.

The study is intended as an exploratory modelling exercise rather than a fully calibrated traffic assessment. The analysis focuses on identifying operational constraints, evaluating responses to changes in demand and control, and recognising the limitations of the available data and simulation setup.

2 CHAPTER

Study Area and Data Sources

2.1 Study Location

The study focuses on the signalised R40–N430 intersection in the Neuseplein–Tolpoort area of Ghent, Belgium. The model represents the main junction and the adjacent network elements needed to simulate vehicle, bicycle and tram movements through the study area.

For consistency with the traffic count dataset, the five approaches are identified as follows:

Table 1 Traffic-count approach identifiers.

Approach	Location
A	R40, Sassekaai side
B	Neuseplein
C	N430
D	Blaisantvest
E	R40, N458 side

These approach identifiers refer to the traffic count locations and should not be confused with the signal-group identifiers used in the official signal timing plan.

2.2 Traffic Count Data

The principal demand dataset consists of traffic counts provided by AWV for 9 February 2023. The model uses the afternoon peak-hour observations from 16:00 to 17:00.

The dataset includes turning movements for motor vehicles and bicycles. These observations were used to define the traffic inputs and routing proportions in the Vissim network.

The recorded motor vehicle demand totals 3,059 vehicles per hour, while the bicycle demand totals 349 bicycles per hour.

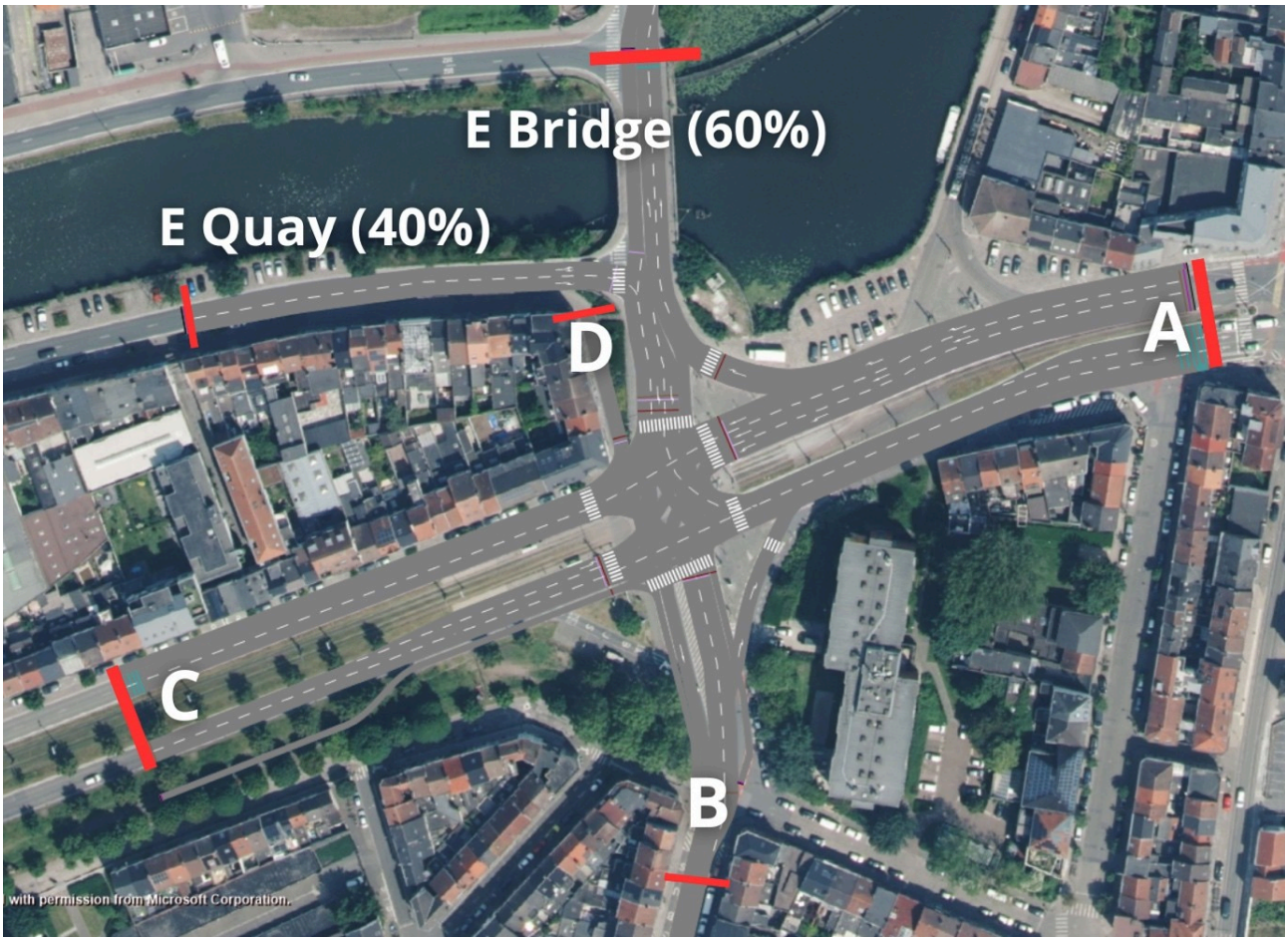


Figure 1 Study area and traffic-count approach definitions (A–E) at the R40–N430 intersection in Ghent. Approach E is represented by two modelled entry points, bridge and quay, using an assumed 60/40 demand distribution.

Table 2 Observed PM peak-hour demand by approach (16:00–17:00, 9 February 2023).

Approach	Motor vehicles (veh/h)	Bicycles (bikes/h)
A	1,011	97
B	461	75
C	685	108
D	2	6
E	900	63
Total	**3,059**	**349**

The observed turning-movement matrices provide the basis for assigning traffic to individual movements. Where the available counts do not distinguish between physical entry points, additional assumptions are required to distribute demand within the model. These are documented in Section 3.

2.3 Signal Control Data

The baseline traffic signal control was based on an official AWV signal timing plan (V-plan) for the study intersection.

The main intersection (K1) was modelled using the fixed-time PM signal programme, with a cycle length of 105 seconds. The Tr = 0 branch of the official plan was used as the baseline reference.

The signal-group timings, their implementation in Vissim and any necessary approximations are described in Section 4.

2.4 Scope of the Simulation

The analysis uses the observed one-hour traffic demand as the basis for simulation inputs. Individual simulation runs are limited to 600 seconds, reflecting the constraints of the PTV Vissim Student Version used for the project.

Motor vehicles and bicycles are explicitly represented. Pedestrian movements were excluded to keep the analysis focused on motor vehicle traffic, bicycle interactions and tram operations.

Tram operations are introduced in a separate scenario rather than included in the baseline. This allows the effects of the additional tram movements and simplified priority control to be examined under a defined motor traffic demand level.

The simulation is therefore designed for comparative scenario analysis within a simplified network representation, rather than as a complete reproduction of all traffic operations at the intersection.

3 CHAPTER

Network Reconstruction and Demand Modelling

3.1 Network Reconstruction

The intersection was reconstructed in PTV Vissim 2026 using the available road layout and supporting AWV documentation. The model includes the main signalised junction (K1) and the adjoining road sections required to represent the five traffic-count approaches (A–E).

Road geometry was represented using Vissim links and connectors, incorporating the principal traffic lanes, turning movements and relevant bicycle infrastructure. Dedicated bicycle links were included where separate cycling facilities were represented, while shared-road sections were used for movements following the carriageway.

The network was developed to capture traffic interactions within the study area rather than to reproduce the entire surrounding road network. Signal control and conflict-priority relationships are discussed separately in Section 4.

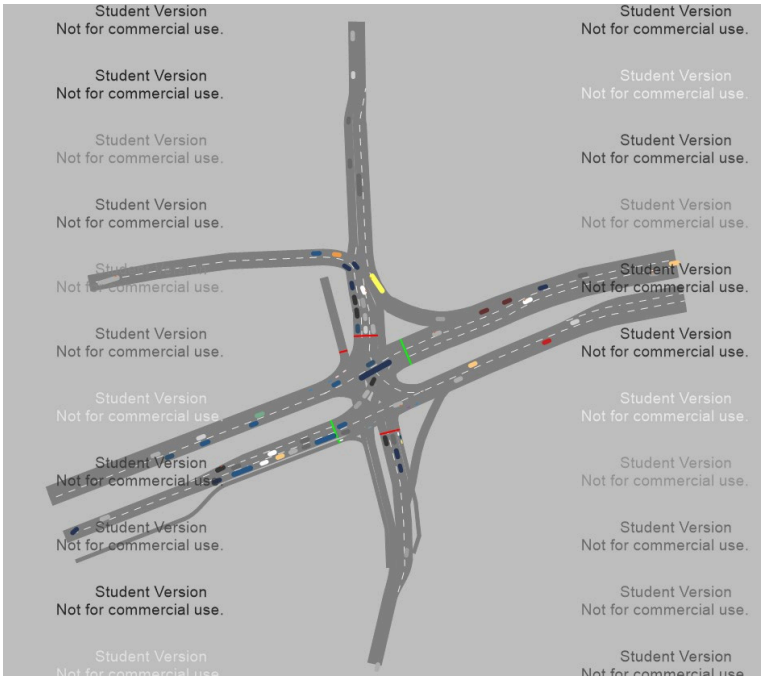


Figure 2 Reconstructed Vissim network of the K1 intersection and the adjoining road sections.

3.2 Motor Vehicle Demand and Routing

Motor vehicle demand was derived from the AWV turning-movement counts collected on 9 February 2023, using the 16:00–17:00 observation period.

The observed demand of 3,059 vehicles per hour was assigned to the corresponding approaches. Vehicle routing was implemented through static routing decisions, using the observed turning volumes to determine the relative proportions of movements.

Approach E required an additional modelling decision. The observed count identifies a combined approach demand of 900 vehicles per hour, whereas the Vissim network represents two upstream entry points: the bridge and the quay/Gasmeterlaan side.

As the available counts do not distinguish between these origins, a 60/40 distribution was assumed:

Table 3 Assumed distribution of approach E motor vehicle demand between the two modelled entry points.

Entry point	Assumed share	Motor vehicle input
E Bridge	60%	540 veh/h
E Quay	40%	360 veh/h

Both entry points were assigned the same observed turning proportions for approach E. The 60/40 distribution is a modelling assumption rather than an observed traffic characteristic.

The vehicle composition was simplified to 96% passenger cars and 4% heavy goods vehicles (HGVs). This composition was adopted for the simulation and was not independently established from the vehicle-category breakdown in the observed counts.

3.3 Bicycle Demand and Representation

Bicycle demand was taken from the same AWV count dataset, with 349 bicycles recorded during the analysed PM hour.

Bicycles were represented as a separate vehicle composition, with dedicated vehicle inputs and static routing decisions for the principal observed movements. The network includes bicycle-specific links and connectors, together with sections where bicycle movements share the carriageway with motor vehicles.

For approach E, the observed bicycle demand of 63 bicycles per hour was divided between the two modelled upstream entry points using the same assumed 60/40 distribution. The resulting inputs were rounded to 38 bicycles per hour from the bridge and 25 from the quay.

Bicycle demand was maintained at the observed level across the motor-traffic demand scenarios. This allowed the effects of increased motor vehicle volumes to be examined without simultaneously changing bicycle inputs.

3.4 Modelling Scope and Assumptions

The network representation combines observed traffic demand with assumptions required to translate the available data into a microscopic simulation model.

The principal assumptions concern the upstream demand distribution at approach E and the simplified motor vehicle composition. Motor vehicles and bicycles were explicitly modelled, while pedestrian flows were excluded to keep the analysis focused on motor traffic, bicycle interactions and tram operations. Pedestrian crossing markings were included in selected visualisations to represent the intersection layout, but they do not indicate simulated pedestrian movements. Tram operations are introduced in a separate scenario and are discussed in Section 7. These modelling choices define the scope of the analysis and are considered when interpreting the scenario results.

4 CHAPTER

Signal Control and Conflict Modelling

4.1 Baseline Signal Control

Signal control at the main intersection (K1) was implemented using the official AWV signal plan (V010049 v17) as a reference. The model uses the 105-second fixed-time programme corresponding to the Tr = 0 operating condition.

The signal programme was reconstructed in Vissim through individual signal groups controlling the relevant motor vehicle and bicycle movements. The principal groups include A, B, G and H for motor traffic, together with the lower-case bicycle signal groups a, b and g.

Selected green intervals within the 105-second cycle are summarised below.

Table 4 Selected green intervals within the 105-second K1 cycle.

Signal group	Green interval (s)
A	81–105 and 0–12
B	20–74
G	20–47
a	20–42
b	81–105
g	81–105

The signal-group identifiers refer to the AWV control plan and are distinct from the approach labels used in the traffic-count dataset.

The original signal plan also includes special signal states. These were represented in Vissim where applicable, although the flashing-amber operation of group H required an approximation using off and amber states.

The same K1 fixed-time programme was retained in the baseline and motor-demand stress-test scenarios.

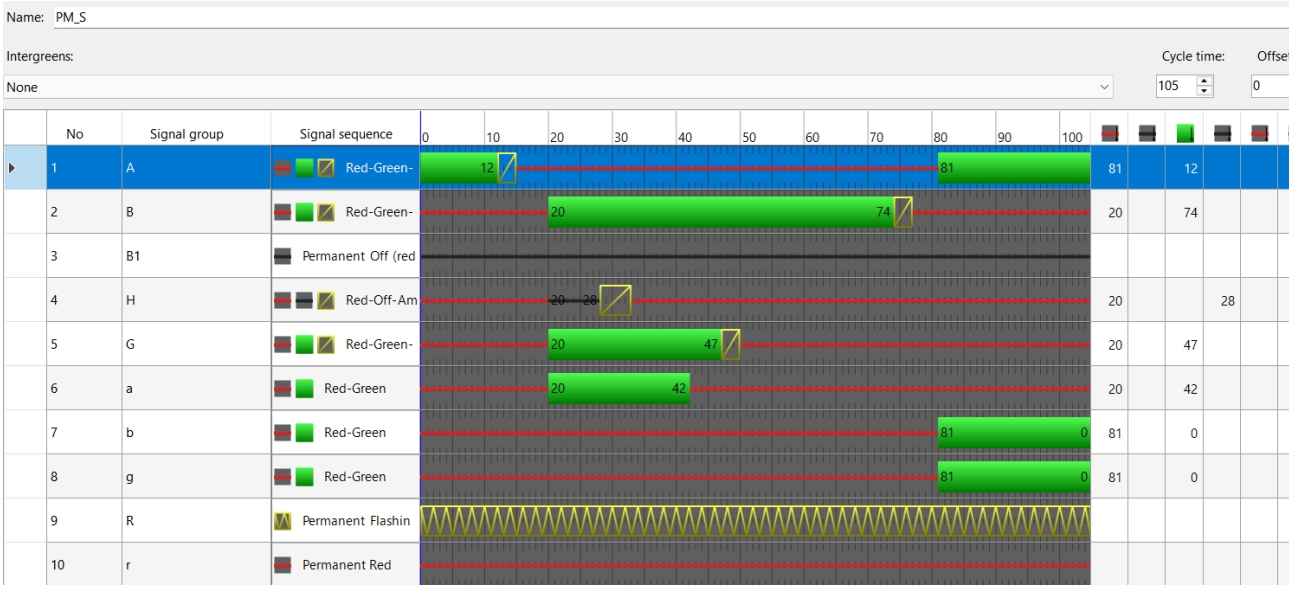


Figure 3 Fixed-time signal programme implemented at K1 in PTV Vissim (105-second cycle, PM_S), based on AWV V-plan V010049 v17.

4.2 Conflict and Priority Modelling

Conflict Areas were used to define priority relationships between intersecting vehicle movements.

Where traffic streams were separated by signal control, the corresponding Conflict Areas could be treated as passive. For movements requiring additional priority control, active Conflict Areas were configured to represent yielding behaviour.

Particular attention was given to interactions between turning motor vehicles and bicycles. Where appropriate, permissive turning movements were required to yield to conflicting bicycle movements.

The conflict configuration was developed to represent the principal interactions within the modelled network. It was not intended as an exact reconstruction of every priority relationship observed in the field.

4.3 Control Strategy Across Scenarios

The original fixed-time K1 controller was retained when motor vehicle demand was increased by 10% and 20%.

In the tram scenario, the 105-second programme also remained in operation. Tram priority was implemented through additional detector-actuated controllers and signal heads, rather than by replacing the original K1 controller.

The tram-priority configuration, detector arrangement and control sequence are described separately in Section 7.

5 CHAPTER

Baseline Assessment

5.1 Assessment Approach

The baseline model was assessed by comparing simulated queue lengths with field observations provided by AWV for 9 February 2023.

The observed dataset reports maximum queue lengths at 15-minute intervals, measured by lane and rounded to the nearest 5 metres. For each approach, the range of observed maximum queues during the 16:00–17:00 period was used as a reference.

The Vissim baseline was evaluated over ten simulation runs, each lasting 600 seconds. This simulation duration was determined by the PTV Vissim Student Version, which limits individual simulation runs to a maximum of 600 seconds (10 minutes). Two queue measures were considered: mean queue length (QLen) and maximum queue length (QLenMax). The reported simulation results represent averages across the ten runs, with standard deviations used to describe variability.

Because the simulation duration was shorter than the field observation intervals, and the queue measurements differed in their aggregation, the comparison was treated as a baseline plausibility check rather than a formal statistical calibration.

5.2 Baseline Queue Results

Table 5 Observed queue ranges and simulated baseline queue lengths by approach.

Approach	Observed queue range (m)	Mean QLen (m)	Mean QLenMax (m)	SD QLenMax (m)
A	65–100	19.1	89.2	30.3
B	35–70	16.7	57.0	8.3
C	55–100	15.7	55.4	11.4
D	0	0.2	4.0	3.0
E	30–70	20.0	49.0	3.8

Source: AWV queue observations (9 February 2023) and Vissim baseline results (10 simulation runs).

The simulated mean maximum queue lengths for approaches A, B, C and E fall within the corresponding observed ranges. Approach D, which carries very limited motor traffic, shows a small simulated maximum queue of approximately 4 m compared with zero in the field observations.

Approach A exhibits the greatest variability, with a standard deviation of approximately 30 m and individual run maxima ranging from 45 to 131 m. Figure 4 compares the observed queue ranges with the simulated mean maximum queue lengths in the baseline scenario.

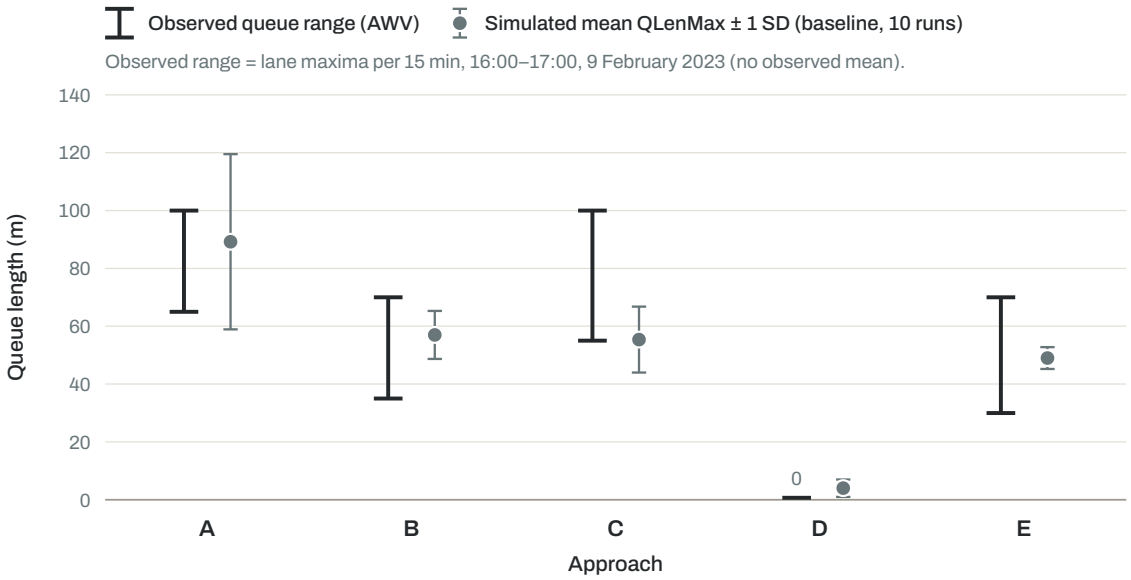


Figure 4 Observed AWV queue ranges (dark vertical intervals: range of lane maxima per 15 minutes, 16:00–17:00, 9 February 2023; no observed mean is shown) compared with the simulated baseline mean maximum queue length (QLenMax; grey points with error bars showing ±1 SD across ten Vissim runs).

For approach E, the baseline comparison uses the queue counter near the intersection. Additional counters on the bridge and quay branches recorded mean maximum queues of 72.4 m and 57.2 m, respectively. As separate field queue observations were unavailable for these branches, their results could not be independently assessed against observed ranges.

5.3 Interpretation

The comparison indicates that the model reproduces the approximate magnitude of observed queues on the principal approaches. However, the variability on approach A and the differences between field and simulation measurement periods limit the strength of this assessment.

The 600-second simulation horizon also limits the ability to assess queue development over an extended peak period. The baseline was therefore considered sufficiently plausible for exploratory demand stress testing and scenario comparison, without claiming that the model had undergone comprehensive field calibration or validation.

6 CHAPTER

Motor Traffic Demand Stress Testing

6.1 Scenario Design

Two demand stress tests were conducted to examine how the intersection responds to increased motor vehicle volumes.

Motor vehicle demand was increased by 10% and 20% relative to the observed PM-hour counts, while bicycle demand remained fixed at 349 bicycles per hour. The baseline fixed-time signal programme was retained.

Table 6 Motor vehicle demand in the baseline and stress-test scenarios.

Scenario	Motor vehicle demand (veh/h)	Increase
Baseline	3,059.0	–
Demand +10%	3,364.9	+10%
Demand +20%	3,670.8	+20%

Each scenario was evaluated using ten 600-second simulation runs. Queue length indicators were compared across the scenarios to examine the response to increasing traffic demand.

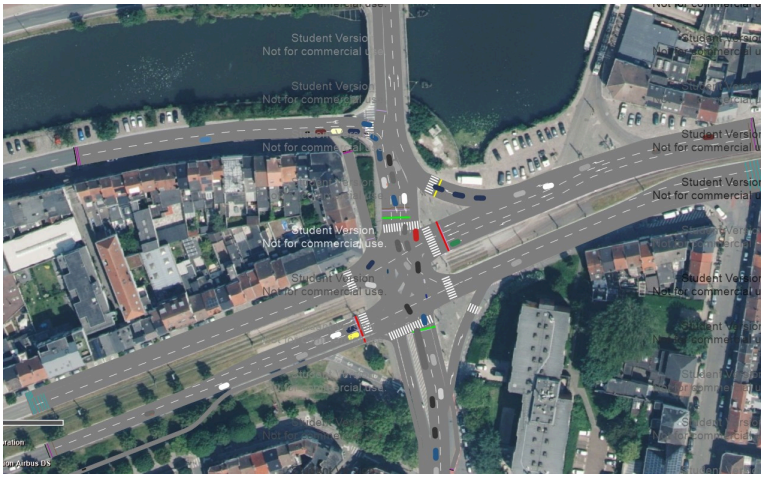


Figure 5 Baseline traffic simulation at the observed motor vehicle demand (3,059 veh/h), showing vehicle movements through the reconstructed K1 intersection.

6.2 Queue Performance

The results show a substantial increase in queue lengths on several approaches as motor vehicle demand rises.

Table 7 Mean maximum queue length (QLenMax) by queue counter and motor-demand scenario, in metres.

Queue counter		● Baseline		■ +10%		▲ +20%	
		MEAN	SD	MEAN	SD	MEAN	SD
A	R40, Sassekaai side	89.2	± 30.3	124.4	± 19.5	142.8	± 27.9
B	Neuseplein	57.0	± 8.3	67.0	± 11.3	83.4	± 17.7
C	N430	55.4	± 11.4	62.0	± 12.4	77.8	± 17.8
D	Blaisantvest	4.0	± 3.0	4.0	± 3.0	3.7	± 2.7
E	R40, N458 side	49.0	± 3.8	48.8	± 3.5	49.4	± 3.8
APPROACH E — UPSTREAM BRANCHES							
E Quay	upstream branch, 40 % of E	57.2	± 24.3	78.1	± 30.3	98.0	± 21.6
E Bridge	upstream branch, 60 % of E	72.4	± 23.9	88.5	± 21.5	92.4	± 18.8

Mean QLenMax ± sample standard deviation (SD), based on ten 600-second simulation runs per scenario. Values in metres.

Approach A shows the largest increase in mean maximum queue length among the main approach counters, from 89.2 m in the baseline to 124.4 m under +10% demand and 142.8 m under +20% demand. These results indicate more pronounced queueing in the increased-demand scenarios, although differences between the saved network configurations limit direct attribution of the changes to traffic demand alone.

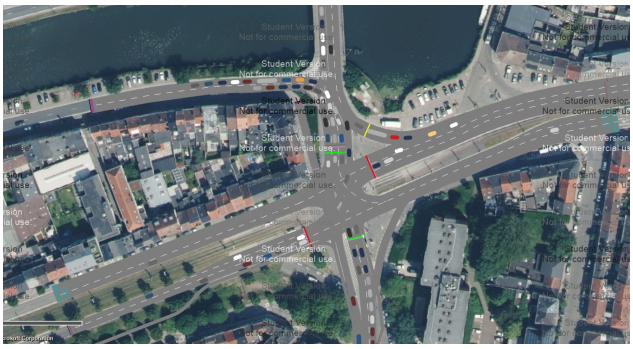


Figure 6 Illustrative traffic conditions at the K1 intersection under a 10% increase in motor vehicle demand (3,364.9 veh/h).



Figure 7 Illustrative traffic conditions at the K1 intersection under a 20% increase in motor vehicle demand (3,670.8 veh/h).

The mean queue length on approach A also increases substantially, from 19.1 m in the baseline to 42.6 m and 55.6 m under the two demand scenarios.

Approaches B and C show a similar upward trend, although the changes are less pronounced.

The additional counters on the two upstream branches of approach E also indicate growing congestion. At E Quay, mean QLenMax increases from 57.2 m to 98.0 m under +20% demand. At E Bridge, it increases from 72.4 m to 92.4 m. Approach E is represented by a queue counter near the intersection and two additional counters on its upstream bridge and quay branches. The branch results are reported separately and depend on the assumed 60/40 demand distribution.

The simulation snapshots provide visual examples of traffic conditions in individual runs. Quantitative comparisons are based on the results from ten runs per scenario. Figure 8 summarises the change in mean maximum queue lengths across the three motor-demand scenarios.

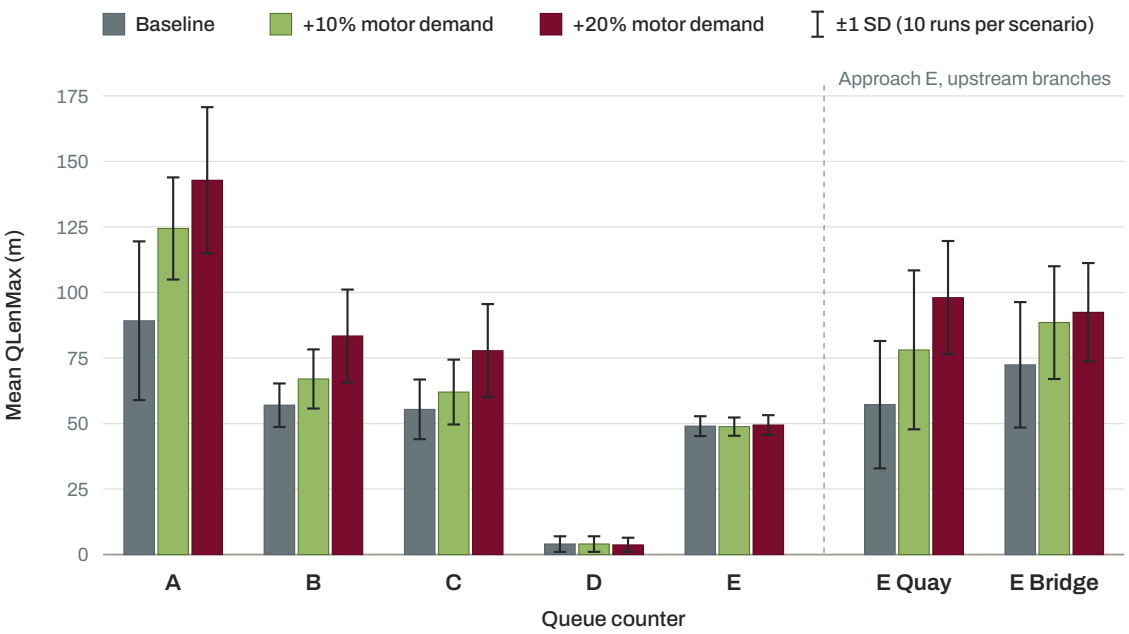


Figure 8 Mean maximum queue lengths (QLenMax) under baseline, +10% and +20% motor vehicle demand. Error bars show ±1 SD across ten runs.

6.3 Interpretation

The demand stress tests indicate increasing queueing pressure under higher motor vehicle volumes, particularly on approach A. Higher mean maximum queue lengths were also recorded on approaches B and C and on the two upstream branches of approach E.

The +20% scenario produced the highest mean maximum queue lengths on several approaches, suggesting that the modelled intersection may be vulnerable to further demand growth. However, the differences between saved network configurations and the limited upstream network extent should be considered when interpreting these changes.

The short simulation period also restricts the assessment of sustained queue accumulation and longer-term oversaturation. The results therefore provide an exploratory indication of congestion pressure rather than a definitive estimate of intersection capacity. Further methodological limitations are discussed in Section 8.2.

7 CHAPTER

Tram Operations and Priority Modelling

7.1 Tram Scenario Design

A separate scenario was developed to examine how tram operations and priority control affect intersection performance.

The tram scenario was tested under +10% motor vehicle demand, with bicycle demand maintained at 349 bicycles per hour. This demand level was selected because the +20% scenario already exhibited substantial congestion, making the additional effects of tram operation more difficult to interpret.

Tram movements were represented using three Vissim public transport (PT) lines, reflecting different trajectories through the intersection and the adjoining loop. Two tram stops were included, with a fixed dwell-time distribution of 20 seconds. The tram desired-speed distribution was set to 30–35 km/h.

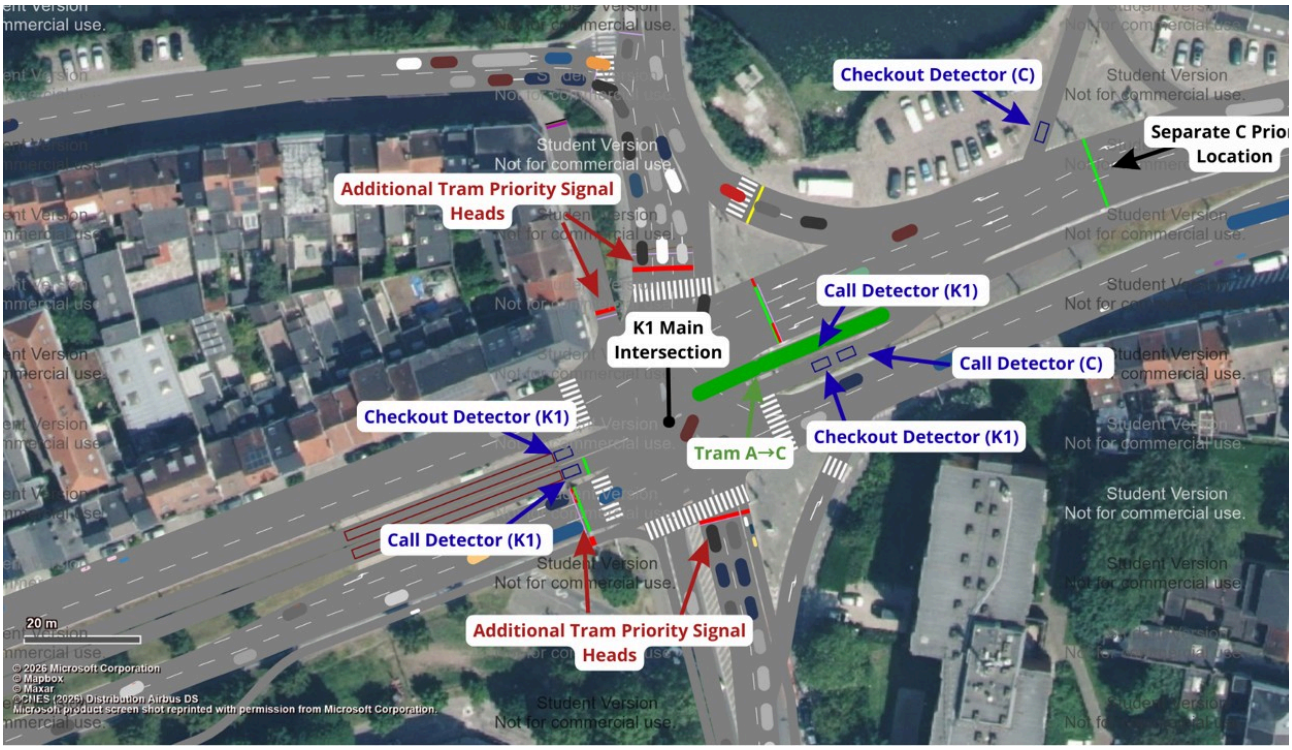


Figure 9 Tram trajectories and priority-control locations within the Vissim network.

7.2 Tram Priority Implementation

Tram priority was implemented using two additional detector-actuated Railway Crossing controllers. The original 105-second fixed-time programme remained active at K1, while supplementary signal heads temporarily restricted movements conflicting with approaching trams. This arrangement represents an alternative tram-priority strategy rather than an exact reproduction of the conditional tram-control sequence documented in the official AWV V-plan.

The priority-control sequence was visually tested in Vissim, including detector activation, clearance of conflicting movements, tram passage and checkout release. These checks supported the functional operation of the implemented strategy, but did not establish its equivalence to the official AWV tram-control programme.

The two additional controllers serve different conflict locations:

Controller	Vissim name	Function
SC3	K1_TRAM_PRIORITY	Controls conflicting movements at the main K1 intersection.
SC2	TRAM_PRIORITY	Provides separate priority control at the downstream signal-group-C conflict location.

Call detectors identify approaching trams and trigger the closure of conflicting movements. Following a clearance interval, the tram is permitted to proceed. Checkout detectors release the priority request after the tram has passed.

The K1 controller uses a 10-second clearance time, while the separate C controller uses 5 seconds.

Additional signal heads were assigned to the relevant conflicting motor vehicle and bicycle movements. These operate alongside the original fixed-time signals rather than replacing them.

The tram movement from the C-side towards A uses sequential priority requests at K1 and the separate C-controlled location. The third PT line represents a movement connecting to the external loop through shared traffic lanes and does not use a dedicated tram-priority request.

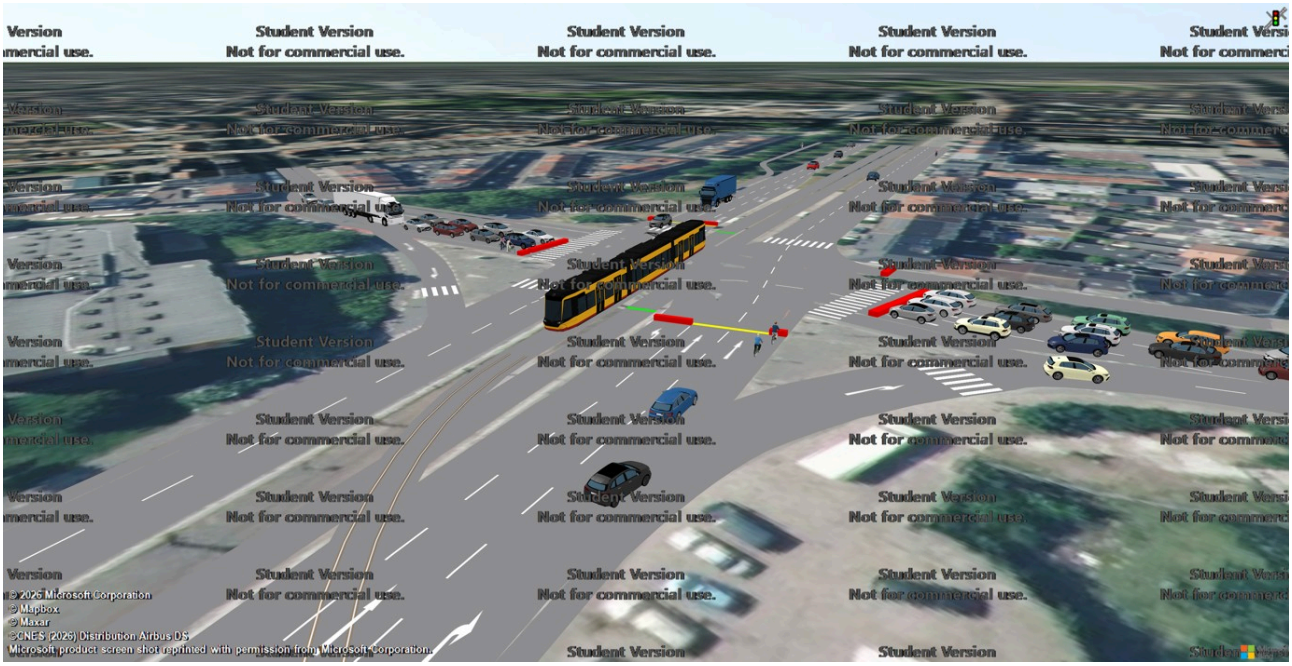


Figure 10 3D visualisation of tram movement through the modelled K1 intersection under +10% motor vehicle demand with detector-actuated tram-priority control.

7.3 Scenario Comparison

The +10% demand scenarios with and without tram operations were compared using ten 600-second simulation runs each.

Table 8 Mean maximum queue length (QLenMax) under +10% motor vehicle demand without and with tram operations, in metres.

Queue counter		+10% without tram MEAN	+10% without tram SD	+10% with tram MEAN	+10% with tram SD	Difference MEAN
A	R40, Sassekaai side	124.4	± 19.5	129.6	± 47.1	+5.2
B	Neuseplein	67.0	± 11.3	68.5	± 9.7	+1.5
C	N430	62.0	± 12.4	63.2	± 13.1	+1.2
D	Blaisantvest	4.0	± 3.0	4.0	± 3.0	0.0
E	R40, N458 side	48.8	± 3.5	49.4	± 3.5	+0.6
APPROACH E — UPSTREAM BRANCHES						
E Quay	upstream branch, 40 % of E	78.1	± 30.3	76.4	± 24.4	–1.7
E Bridge	upstream branch, 60 % of E	88.5	± 21.5	89.8	± 27.1	+1.3

First two column groups report mean QLenMax ± sample SD across ten runs. Difference = mean with tram – mean without tram; it does not imply a statistically established effect.

The mean maximum queue lengths changed relatively little on most approaches following the introduction of tram operations and priority control.

Approach A showed a modest increase in mean QLenMax, from 124.4 m to 129.6 m. However, its run-to-run variability increased substantially, with the standard deviation rising from 19.5 m to 47.1 m.

7.4 Interpretation

Within the 600-second simulation period, introducing tram operations with detector-actuated priority was associated with relatively small changes in mean maximum queues on most measured approaches.

The greater variability on approach A indicates that individual simulation outcomes can differ considerably, making the small difference in mean QLenMax insufficient to establish a consistent deterioration.

The results should not be interpreted as evidence that tram priority has no effect on traffic performance. The analysis is limited to a short simulation period, selected queue measures and the implemented priority strategy. Tram travel-time or reliability benefits were not separately evaluated.

The scenario therefore provides an exploratory assessment of how modelled tram operations interact with an existing signalised intersection under increased motor vehicle demand.

8 CHAPTER

Discussion and Model Limitations

8.1 Main Findings

The scenario results provide insight into how the modelled intersection responds to increased motor vehicle demand and tram-priority operation.

The baseline assessment showed that simulated maximum queue lengths were broadly consistent with the available AWV observations. Although this does not constitute comprehensive model validation, it provided a reasonable starting point for exploratory scenario analysis.

The increased-demand scenarios showed higher mean maximum queue lengths on several approaches, particularly approach A. Increases were also recorded on approaches B and C and on the upstream branches of approach E. These results suggest increasing queueing pressure under the modelled conditions. However, differences between the saved network configurations and the limited upstream network extent restrict the extent to which these changes can be attributed exclusively to increased demand.

The introduction of tram operations with detector-actuated priority under +10% motor vehicle demand produced relatively small changes in mean maximum queue lengths on most approaches. However, increased variability on approach A indicates that individual simulation outcomes may differ substantially. The results do not establish whether the priority strategy improves tram travel times or overall network performance, as these outcomes were not directly evaluated.

8.2 Model Limitations

Several limitations should be considered when interpreting the simulation results.

Each simulation run was limited to 600 seconds by the PTV Vissim Student Version. Queue measurements also began with an initially empty network, without a separate warm-up period. Although ten runs were performed per scenario, the results cannot fully represent queue development throughout the PM peak hour. Furthermore, the available field observations differed from the simulated queue measurements in their time intervals and measurement methods. The baseline comparison was therefore limited to an assessment of approximate queue magnitudes rather than formal calibration.

The model also required assumptions where detailed data were unavailable. In particular, the demand from approach E was distributed between the bridge and quay using an assumed 60/40 split, while vehicle composition and selected bicycle parameters were simplified. Differences between the saved baseline and increased-demand network configurations, including the upstream extent of approach A, limit the extent to which changes in queue length can be attributed exclusively to increased demand. The limited upstream network extent may also affect recorded maximum queue lengths, particularly on approach A and the two branches of approach E.

Bicycle movements were explicitly represented, but pedestrian flows were not simulated. The tram scenario used an alternative detector-actuated priority strategy rather than reproducing the conditional tram-control logic documented in the official AWV signal plan. Consequently, the tram results should be interpreted as an exploratory assessment of the implemented control strategy, not as a validated representation of current traffic operations.

8.3 Implications for Interpretation

The model provides a useful basis for examining traffic interactions and comparing operational responses under different demand and control conditions. However, the results should not be used directly to support infrastructure or signal-control decisions without further validation.

A more comprehensive assessment would require longer simulation periods, additional field observations, consistent network configurations across scenarios, and broader performance indicators such as vehicle delay, throughput and public transport travel times.

Within its defined scope, the project illustrates a practical microscopic modelling workflow, from reconstructing a real intersection and coding observed demand to testing scenarios and critically interpreting simulation results.

9 CHAPTER

Conclusions

This project examined traffic operations at the R40–N430 intersection in Ghent through microscopic simulation in PTV Vissim, using observed AWV traffic counts and an official signal plan.

The baseline model produced queue lengths broadly comparable to the available field observations, providing a starting point for exploratory scenario analysis. Increased-demand scenarios showed higher mean maximum queue lengths on several approaches, particularly approach A, where the value reached 142.8 m under +20% demand. These findings suggest increasing congestion pressure under the modelled conditions, although differences between network configurations and the limited simulation period restrict the strength of this interpretation.

A separate scenario incorporated tram operations and detector-actuated priority control while retaining the original fixed-time signal programme. Under +10% motor vehicle demand, introducing tram operations resulted in relatively small changes in mean maximum queues on most approaches, although greater run-to-run variability was observed on approach A.

The findings are subject to the short simulation period and limited field validation data. Nevertheless, the project provides an exploratory assessment of intersection performance under different demand and control conditions.

The work involved reconstructing a real road network, implementing multimodal traffic demand and signal control, developing and testing tram-priority logic, and interpreting simulation results with explicit consideration of modelling limitations.

APPENDIX

Appendices

A Observed Turning-Movement Matrices

The following matrices present the observed turning movements recorded by AWV during the PM peak hour (16:00–17:00) on 9 February 2023. Rows represent origin approaches and columns represent destination approaches.

Table A1 Observed motor vehicle turning movements (veh/h).

From / To	A	B	C	D	E	Total
A	–	37	394	0	580	1,011
B	62	–	126	0	273	461
C	417	104	–	0	164	685
D	0	0	2	–	0	2
E	567	241	92	0	–	900
Total	**1,046**	**382**	**614**	**0**	**1,017**	**3,059**

Table A2 Observed bicycle turning movements (bicycles/h).

From / To	A	B	C	D	E	Total
A	–	0	77	0	20	97
B	6	–	0	0	69	75
C	99	0	–	0	9	108
D	1	4	1	–	0	6
E	13	41	9	0	–	63
Total	**119**	**45**	**87**	**0**	**98**	**349**

Source: AWV traffic counts, 9 February 2023.
Note: A–E refer to traffic-count approaches, not signal groups. The matrices represent observed demand. The 60/40 split of approach E between bridge and quay is a separate modelling assumption and is not part of the observed matrices.